

# Brandon Loke

+65 8181 0051 | [brandonlokemk@gmail.com](mailto:brandonlokemk@gmail.com) | [LinkedIn](#) | [GitHub](#) | [Website](#)

## EDUCATION

**Nanyang Technological University**

**2020 - 2024**

**Bachelor of Engineering (Computer Science), Honours Graduate**

- Elective Focus: Artificial Intelligence | Data Science and Analytics

## WORK EXPERIENCE

**Front-End Developer Intern**

**Jan 2023 – May 2023**

Republic Power Pte Ltd

Singapore

Tech Stack: JavaScript, React, Cypress, Microsoft Power Automate, UiPath, MQTT, IoT

- Responsible for automating the user flow for the Fundus Camera using **Microsoft Power Automate** and **UiPath**
- Gained experience in setting up and working with various **IoT** products, such as environment sensors, and extracting data from them or their respective cloud platforms
- Developed modular, responsive **React** components to use in dynamic web pages
- Performed End-to-End testing on **React** web applications using **Cypress**, a Front-End testing framework
- Established communication across multiple **IoT** devices by configuring **MQTT** clients and brokers

## PROJECTS

**Osteoporosis Prediction Using Fundus Imaging and Medical Data**

*In collaboration with **Singapore National Healthcare Group (NHG)***

Tech Stack: Python, PyTorch, Deep Learning, Computer Vision, Joint Fusion Models

- Developed a **multi-modal deep learning model** using fundus images and patient metadata to classify individuals into Osteoporosis, Low Bone Density, and Normal categories
- Designed and trained models with **PyTorch** using **transfer learning** and fine-tuning techniques, incorporating **Bayesian hyperparameter tuning** for optimization
- Fused visual and tabular inputs using **joint fusion models**, and applied **ensemble learning** to improve robustness
- Conducted **data augmentation** and **addressed class imbalance** to enhance performance on a limited dataset

**Telegram Job Connect Bot**

Tech Stack: Python, Flask, MySQL, Google Cloud Platform, Telegram API

- Developed a **Telegram bot** using **Python** and **Flask** to bridge job seekers and employers by facilitating job postings and searches in Singapore
- Features multi-account management per Telegram ID, with role-based access (employers/job seekers)
- Implemented a virtual currency system with credits to gate job listings, ensuring spam-free operations
- Hosted on **Google Cloud Run** with **MySQL** database management via **Google Cloud SQL** for reliable performance
- Currently serving 14,000+ members in a Telegram community, improving job search efficiency

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## CO-CURRICULAR ACTIVITIES

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### Nanyang-Technological University

Aug 2021 – July 2023

School of Computer Science and Engineering (SCSE) IT Subcommittee

Tech Stack: TypeScript, React, Next.js, Jest

- Contributed to various projects for NTU's SCSE IT Subcommittee, including development of the merch store and website migration
- Built reusable and type-safe **React** components in **TypeScript** within a **Next.js** framework
- Styled components with Chakra UI CSS framework to ensure consistency and accessibility
- Performed unit testing on components using **Jest**, with a focus on user interaction testing

## SKILLS

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**Languages** | English, Chinese (Basic)

**Programming Languages** | Python, JavaScript, TypeScript, Java

**Frontend** | HTML, CSS, React, Next.js, Tailwind CSS

**Backend** | Node.js, Flask, REST APIs, SQL, MySQL

**Machine Learning** | PyTorch, TensorFlow, Keras, OpenCV

**Testing Frameworks** | Cypress, Jest, Selenium

**Developer Tools** | Git, GitHub, Postman, MQTT, UiPath, Microsoft Power Automate